

Project Report: Import Data using Transform Maps (Spreadsheet)

Team ID: SWTID-2026-4908

Project Name: Import Data using Transform Maps (Spreadsheet)

1. INTRODUCTION

1.1 Project Overview

Organizations frequently need to import structured data from external sources into enterprise platforms like ServiceNow. This project implements an automated, validated data import process using Import Sets and Transform Maps. Employee data containing standard fields—**Employee ID, Name, Email, Department, and Location**—is staged in a temporary Import Set Table (u_employee_import). The data is validated, mapped, and transferred into a custom target table (u_employee_test). By setting the **Employee ID** as a Coalesce field, the system automatically updates existing records or inserts new ones, effectively preventing duplicate entries. Finally, reports and dashboards are generated to provide administrative oversight.

1.2 Purpose

The primary purpose of this project is to establish a reliable, repeatable pipeline for migrating employee spreadsheets into ServiceNow. It ensures data integrity, eliminates manual record duplication through Coalesce logic, and provides real-time visualization of employee metrics by department and location.

2. IDEATION PHASE

2.1 Problem Statement

Manual entry of large employee datasets into enterprise management systems is error-prone, time-consuming, and frequently leads to duplicate records when updating existing employee profiles. Without a staged validation and automated transformation mechanism, inconsistent or redundant data degrades database accuracy and hinders reporting.

2.2 Empathy Map Canvas

- **Says:** "Manually adding or updating hundreds of employee records takes hours, and we keep getting duplicate accounts."
- **Thinks:** "There should be an automated way to upload spreadsheets where existing records update automatically and new employees are created seamlessly."
- **Does:** Uploads bulk Excel files periodically and spends extra time manually auditing data quality and cleaning up duplicates.
- **Feels:** Frustrated by repetitive manual uploads and worried about data inconsistency affecting department analytics.

2.3 Brainstorming

To solve the issue of manual entry and duplicate creation, several ideas were evaluated:

1. **Direct Scripted Import:** Writing custom background scripts to parse raw CSV files directly into target tables. *(Rejected: High maintenance and lack of low-code reusability).*
2. **Standard Import Sets with Manual Auditing:** Using ServiceNow Import Sets but checking for duplicates manually post-import. *(Rejected: Fails to eliminate manual overhead).*
3. **Transform Maps with Coalesce Rules & Dashboard Integration (Chosen):** Leveraging ServiceNow Import Set Tables as staging areas, auto-mapping matching fields, configuring Employee ID as a Coalesce key to handle updates/inserts automatically, and displaying analytical charts on a dashboard.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

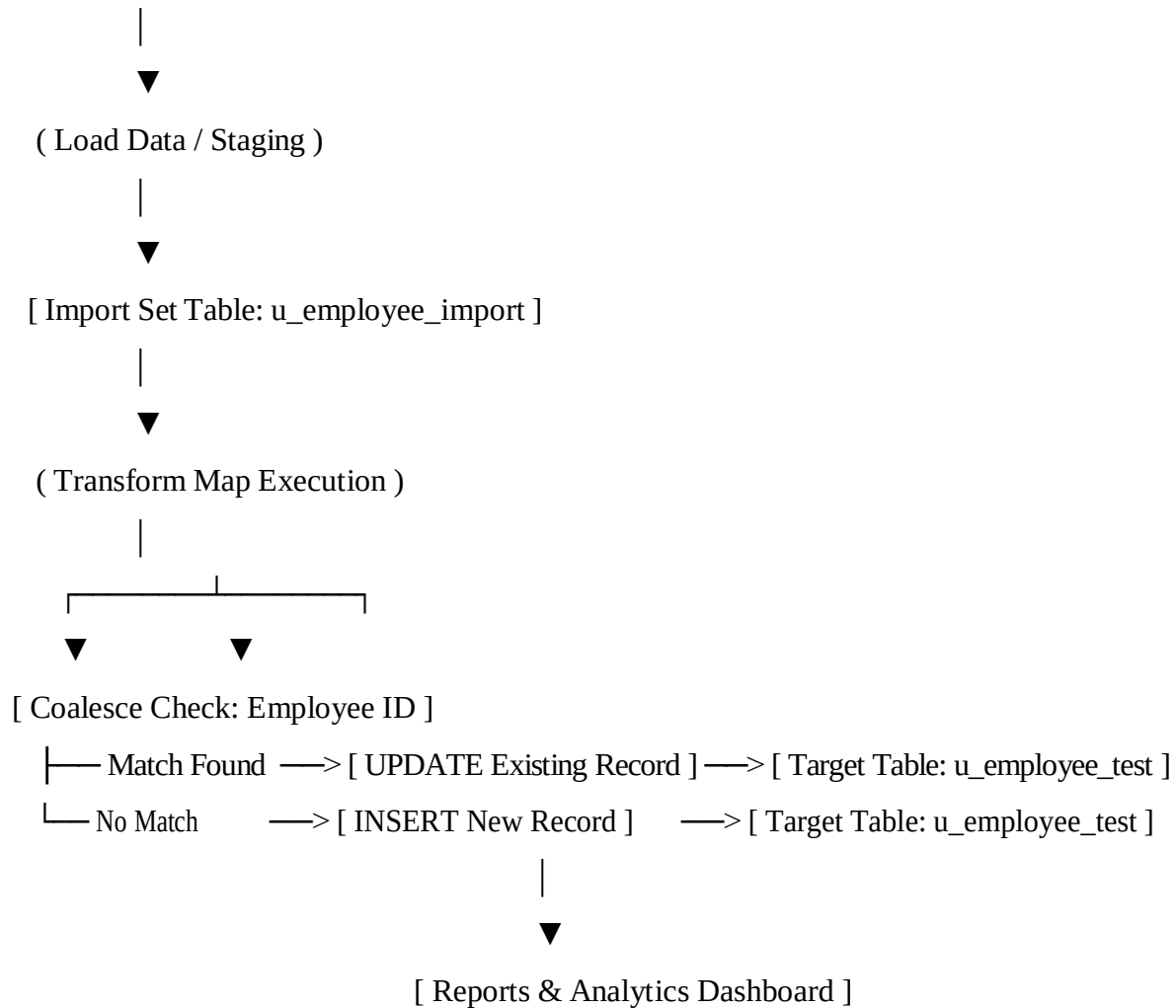
1. **Spreadsheet Preparation:** User creates/downloads employee data in standard .xlsx format.
2. **Data Staging:** User navigates to **System Import Set > Load Data** and uploads the spreadsheet into the staging table u_employee_import.
3. **Field Mapping & Transformation:** System maps source columns to u_employee_test target fields and executes the transformation.
4. **Validation & Coalesce Enforcement:** System checks Employee ID; existing IDs trigger record updates while new IDs create new rows.
5. **Analytics Review:** Administrator opens the Employee Analytics Dashboards page to monitor departmental and location distributions.

3.2 Solution Requirement

- **Functional Requirements:**
 - Custom target table u_employee_test containing Employee ID, Employee Name, Email, Department, and Location string fields.
 - Staging import set table u_employee_import created via file upload.
 - Transform map Sample Spreadsheet Import with mapped source and target fields.
 - Coalesce rule enabled on Employee ID.
 - 3 Reports (Pie charts by Department & Location, and a List view) integrated into a single dashboard.
- **Non-Functional Requirements:**
 - **Data Integrity:** Zero duplicate records upon repeated spreadsheet uploads.
 - **Usability:** Simplified file upload interface utilizing native ServiceNow Import Set utilities.

3.3 Data Flow Diagram

[Excel / Google Sheet (.xlsx)]



3.4 Technology Stack

- **Platform:** ServiceNow (Developer Instance)
- **Modules Used:** System Import Sets, Transform Maps, Reports, Dashboards (PA_DASHBOARDS)
- **Data Format:** Microsoft Excel (.xlsx) / Google Spreadsheets
- **Platform Tools:** Form Layout Configuration, Personalize List Columns

4. PROJECT DESIGN

4.1 Problem Solution Fit

By matching spreadsheet headers directly to ServiceNow target table fields, the transform map automates data ingestion. Configuring **Employee ID** as Coalesce solves the duplicate record issue directly at the database staging level before permanent rows are modified or added.

4.2 Proposed Solution

The solution implements a 4-milestone architecture:

1. **Target Table Construction:** Creation of custom table u_employee_test with explicit field types.
2. **Staging & Mapping Setup:** Creation of u_employee_import staging table and Sample Spreadsheet Import map using **Auto Map Matching Fields**.
3. **Data Validation & Deduplication:** Setting Coalesce = True on Employee ID to enforce insert/update logic.
4. **Reporting:** Building visual pie charts (Employees by Department, Employees by Location) and adding them to Employee Analytics Dashboards.

4.3 Solution Architecture

The architecture separates staging operations from production target records:

- **Staging Layer:** Temporary storage (u_employee_import) receives raw spreadsheet contents.
- **Transformation Layer:** Business rules and mapping scripts transform incoming data types and match target attributes.
- **Target Layer:** Production table (u_employee_test) stores clean, deduplicated employee records.
- **Visualization Layer:** ServiceNow Reporting Engine projects aggregated views onto Performance Analytics Dashboards.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Milestone / Task	Description	Assigned To	Duration
Milestone 1: Creation Of Spreadsheet And Table	Create sample .xlsx employee dataset; configure custom target table u_employee_test and form layout.	Afrin A	40m
Milestone 2: Creation Of Import Set Table And Transform Map	Upload spreadsheet to create staging table u_employee_import; define transform map and auto-map fields.	Archana N	40m
Milestone 3: Transform Data & Validate And Enable Coalesce	Validate initial import; set Coalesce on Employee ID; test updates, inserts, and duplicate ignore logic.	Esakkithal E / Nardhana T K	10h
Milestone 4: Creation of reports & dashboards	Build 3 reports (Department, Location, List); configure Employee Analytics Dashboards.	Shifana M A	40m

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The data transformation and deduplication mechanics were tested under various upload conditions:

- **Initial Data Import Test:**
 - Uploaded initial dataset containing employee records.

- **Result:** All records correctly transformed and inserted into u_employee_test.
- **Update & Insert Test (Coalesce Active):**
 - Uploaded a modified Excel sheet containing 4 total rows: 2 existing records with modified details (email/name) and 2 brand-new Employee IDs.
 - **Result:** Transform History confirmed **2 Inserted** and **2 Updated**. Target table reflected new emails/names without producing duplicate rows.
- **Duplicate Upload Test (Ignored Records):**
 - Re-uploaded the identical 4-row spreadsheet.
 - **Result:** Transform History confirmed **0 Inserted**, **0 Updated**, and **4 Ignored**. Database maintained complete integrity.

7. RESULTS

7.1 Output Screenshots

1. **Spreadsheet Preparation:** Sample Google/Excel spreadsheet formatted with columns (Employee ID, Name, Email, Department, Location).
2. **Target Table Layout:** u_employee_test table configured in Form Layout displaying all custom string fields.
3. **Import Set Staging:** u_employee_import loaded with status "Completed".
4. **Transform Map Configuration:** Sample Spreadsheet Import mapping source fields to target fields with Employee ID Coalesce set to True.
5. **Transform History Log:** Execution status showing **2 Inserted / 2 Updated** on modified load, and **4 Ignored** on re-run.
6. **Analytics Dashboard:** Employee Analytics Dashboards displaying the Department Pie Chart, Location Chart, and Employee List Report widgets.

8. ADVANTAGES & DISADVANTAGES

Advantages

- **Automated Deduplication:** Enforcing Coalesce on Employee ID prevents redundant database entries automatically.
- **Staged Processing:** Staging tables isolate bad or misformatted data before it can corrupt production records.
- **Low-Code Customization:** ServiceNow Transform Maps require minimal coding to map

fields and setup updates.

- **Real-time Analytics:** Direct integration between target tables and dashboards provides instant visibility into workforce metrics.

Disadvantages

- **Header Dependency:** Transformation relies on exact column header matches during auto-mapping.
- **ACL Restrictions:** Creating dashboards via backend forms (PA_DASHBOARDS.FORM) requires administrator override permissions.

9. CONCLUSION

The project successfully automates the importation and validation of employee records into ServiceNow using Import Sets and Transform Maps. By setting Employee ID as the Coalesce key, the system seamlessly updates existing records and inserts new ones while ignoring exact duplicates. The integration of custom pie charts and list views into the Employee Analytics Dashboards delivers a complete end-to-end data pipeline solution.

10. FUTURE SCOPE

- **Scheduled Imports:** Automating periodic spreadsheet ingestion via FTP/SFTP servers using Scheduled Import Sets.
- **Advanced Data Cleaning Scripts:** Adding Transform Field Scripts to format phone numbers, standardize email domain cases, and validate location entries during transformation.
- **Multi-Table Mapping:** Extending transform maps to simultaneously update associated core tables (e.g., sys_user and cmn_department).

11. APPENDIX

Source Code / Scripts

- **Target Table:** u_employee_test
- **Staging Table:** u_employee_import
- **Transform Map Name:** Sample Spreadsheet Import
- **Coalesce Configuration:** Employee ID -> Coalesce = True
- **Dashboard Backend Form:** PA_DASHBOARDS.FORM

Dataset Link

- Sample Dataset: Sample Spreadsheet.xlsx

GitHub & Project Demo Link

- **Team ID:** SWTID-2026-4908
- **Platform Workspace:** myskillwallet.ai Workspace

